

Erwin Quanten

Staff Software Engineer – Flutter, Dart, Firebase, Node.js, AWS

Brussels, Belgium | erwin.quanten.dev@outlook.com | +32460256356 | <https://www.linkedin.com/in/erwin-quanten-b82815a/>

PROFESSIONAL SUMMARY

Staff Flutter Engineer with strong experience building cross-platform mobile applications for **FinTech, logistics, healthcare, SaaS, and enterprise workflow platforms** using **Flutter 2.x–3.x, Dart 2.x–3.x, Firebase, REST APIs, GraphQL, Node.js, AWS, CI/CD, and native Android/iOS integrations**. Specialized in migrating legacy Android/iOS modules into shared Flutter codebases, improving app performance by **25–40%**, reducing crash rates, implementing secure authentication, offline-first flows, push notifications, payment features, and scalable mobile architecture.

SKILLS

- **Mobile: Flutter 2.x–3.x, Dart 2.x–3.x**, Android SDK, Kotlin, Java, iOS integration, Swift basics, MethodChannel, Platform Channels, Flutter Web
- **State Management: Bloc/Cubit**, Provider, Riverpod, GetX, ChangeNotifier, StreamController, RxDart
- **Backend & APIs: Node.js 14–20**, Express.js, NestJS, REST APIs, GraphQL, WebSockets, Firebase Functions
- **Cloud & DevOps: AWS**, Firebase, Google Cloud, Docker, GitHub Actions, Bitrise, Fastlane, Codemagic, CI/CD pipelines
- **Databases**: Firebase Firestore, Realtime Database, SQLite, Hive, PostgreSQL, MongoDB, Redis
- **Mobile Features**: Push Notifications, Firebase Cloud Messaging, Deep Links, In-App Purchases, Stripe, Apple Pay, Google Pay, Maps, Geolocation, Camera, QR Scanner
- **Testing**: Flutter Test, Widget Testing, Integration Testing, Mockito, Patrol, Firebase Crashlytics, Sentry
- **Architecture**: Clean Architecture, MVVM, Modular Architecture, Repository Pattern, Dependency Injection, SOLID, Offline-first Design

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

Houston, TX

- Led the rebuild of a customer-facing **SaaS mobile platform** using **Flutter 3.x, Dart 3.x, Bloc, Firebase, Node.js, and AWS**, replacing fragmented Android and iOS screens with a shared codebase that improved delivery speed by **34%**.
- Designed a **Clean Architecture** mobile foundation with feature modules, repositories, dependency injection, typed API clients, reusable widgets, and environment-based configuration for staging, QA, and production.
- Migrated legacy native mobile features from **Kotlin, Java, and Swift** into Flutter modules while keeping critical platform-specific flows through **MethodChannel** for camera, file upload, biometric login, and secure storage.
- Improved mobile performance by **28%** by reducing unnecessary widget rebuilds, optimizing image caching, splitting large screens into smaller reactive components, and profiling frame rendering with Flutter DevTools.
- Solved recurring production crashes from expired authentication tokens by implementing automatic token refresh, secure session recovery, API retry handling, and Firebase Crashlytics monitoring across release builds.
- Implemented secure login, MFA, biometric authentication, role-based navigation, deep links, and push notification flows using **Firebase Cloud Messaging**, OAuth2, JWT, and encrypted local storage.
- Reduced API-related mobile errors by **31%** by introducing typed response models, centralized Dio interceptors, structured exception handling, and better backend contract validation with Node.js services.
- Built offline-first workflows with **Hive, SQLite, sync queues, and conflict resolution**, allowing users to continue key operations during weak network conditions and sync automatically after reconnection.
- Created reusable mobile design system components for forms, dashboards, modals, tables, and validation states, improving UI consistency across customer and admin mobile experiences.
- Improved release reliability by **26%** by building GitHub Actions, Fastlane, automated signing, Firebase App Distribution, Play Store internal testing, and App Store TestFlight pipelines.
- Mentored Flutter engineers on Bloc patterns, widget testing, performance debugging, secure storage, accessibility, and clean code practices across mobile squads.
- Partnered with product, QA, backend, and DevOps teams to deliver mobile features end-to-end, from API design and UI implementation to release monitoring and production hotfixes.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Built enterprise mobile applications for **FinTech and logistics platforms** using **Flutter 2.x–3.x, Dart 2.x–3.x, Provider, Bloc, Firebase, REST APIs, GraphQL, and AWS-backed services**.
- Migrated multiple Android-first screens into Flutter while preserving native SDK integrations for payments, maps, document scanning, push notifications, and device permission handling.
- Improved user onboarding completion by **22%** by redesigning mobile registration, document upload, form validation, error messaging, and multi-step verification flows.
- Solved a major production issue where users saw blank screens after app updates by fixing route initialization, cached schema mismatch, and stale local database migration handling.
- Implemented payment and wallet-related mobile features using **Stripe, Apple Pay, Google Pay, secure tokenization, webhook-backed backend validation, and transaction status polling**.
- Reduced mobile API latency impact by **27%** by implementing optimistic UI updates, pagination, local caching, request cancellation, and resilient retry strategies.
- Created shared Flutter packages for authentication, networking, analytics, theme management, localization, and reusable UI components across multiple business applications.
- Implemented GraphQL-based dashboard screens with normalized models, pagination, pull-to-refresh, search filters, and error states for operations and finance teams.
- Added integration tests and widget tests for core flows, improving confidence before production releases and reducing manual regression effort for QA teams.
- Worked closely with backend engineers to define API contracts, fix serialization bugs, improve error payloads, and align mobile releases with backend deployment windows.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Developed internal enterprise mobile tools using **Flutter 1.x–2.x, Dart 2.x, Android SDK, Java, Kotlin, REST APIs, Firebase, and Azure/AWS-hosted backend services**.
- Introduced Flutter for cross-platform delivery after comparing native Android and iOS maintenance cost, helping reduce duplicated mobile development effort by **30%**.
- Built warehouse, approval, reporting, and field-service mobile workflows with barcode scanning, camera upload, offline forms, location capture, and push notifications.
- Migrated older Android Java screens into Flutter modules gradually, using hybrid navigation and platform channels to avoid risky full rewrites during active business operations.
- Solved synchronization bugs caused by duplicate offline submissions by adding local transaction IDs, queue locking, backend idempotency checks, and clearer retry states.
- Improved app stability by **24%** through better crash reporting, null-safety preparation, defensive JSON parsing, and stricter release validation across Android device types.
- Implemented role-based access, secure storage, refresh-token handling, and API gateway integration for internal employees and external operational users.
- Created reusable form components, validation rules, localization files, and responsive tablet layouts for teams using different devices in office and field environments.
- Supported CI/CD release automation with GitHub Actions, Firebase App Distribution, Play Store tracks, signed APK/AAB generation, and structured release notes.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Built business mobile and web solutions using **Android Java, Kotlin, early Flutter 1.x, Dart 2.x, REST APIs, Angular, Node.js, SQL Server, and Firebase services**.
- Implemented mobile data-entry workflows for operations teams, including offline draft saving, image upload, barcode scanning, approval routing, and backend synchronization.
- Helped evaluate Flutter as a practical cross-platform option by building proof-of-concept screens for dashboards, forms, navigation, API calls, and reusable UI components.
- Reduced manual reporting time by **21%** by replacing spreadsheet-based field updates with mobile forms connected to backend APIs and centralized reporting dashboards.
- Fixed recurring Android memory issues by optimizing image compression, lifecycle handling, background uploads, and large list rendering in RecyclerView and Flutter ListView screens.

- Integrated Firebase Analytics, Crashlytics, and Cloud Messaging to improve visibility into crashes, adoption, notification delivery, and user behavior across mobile releases.
- Worked with QA and business stakeholders to reproduce device-specific bugs, document edge cases, and ship stable fixes without blocking operational teams.
- Supported backend API improvements in **Node.js and SQL Server**, especially around validation, pagination, file metadata, and mobile-friendly response structures.

Tessengerlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Developed early mobile and enterprise applications using **Java, Android SDK, JavaScript, REST services, SQL Server, HTML, CSS, and backend integration patterns**.
- Built internal Android utilities for field teams, including simple data capture, local storage, image attachment, authentication, and API-based synchronization.
- Improved form submission reliability by **18%** by adding client-side validation, clearer error messages, request retry logic, and safer handling of unstable mobile connections.
- Resolved production issues involving inconsistent API responses, missing required fields, and device-specific layout problems by improving validation and regression testing.
- Collaborated with senior engineers to modernize older web and mobile modules, document reusable patterns, and prepare the codebase for later cross-platform mobile adoption.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009